

Distance Measures II

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Recap

- Last time we covered **Distance Measures I**:
 - Eigenanalysis.
 - Singular value decomposition.
 - Distance measures and their properties.

Introduction

- We have covered some properties of distance measures but very few actual measures so far.
- **There are many distance measures!**
 - *We will cover a small proportion of them in these slides.*
- Depending on your field common distance measures may vary.
 - *Resources to find additional measures will be in the references.*

Things to Consider

- ① What are the **distributional characteristics** of the data?
 - Binary? Categorical? Frequencies? Continuous?

- ② Are **shared absences** meaningful?
 - For example, does a value of zero in a count variable or the absence of an attribute in categorical data have meaning?
 - Data are *symmetric* if a shared absence is meaningful, and *asymmetric* if it is not.

Some Distance Measures

Euclidean Distance

- Euclidean Distance generalizes to many dimensions and can be useful for numeric, binary, and ordinal data.
 - Values may be positive or negative.
 - In R `dist()`. In Python `scipy.spatial.distance_matrix`
- Euclidean Distance (ED) between objects i and h across p dimensions is:

$$ED = \sqrt{\sum_{j=1}^p (a_{hj} - a_{ij})^2} \quad (1)$$

- **The distance depends on the scale of the variables.**
- Generally inappropriate for community data.

Example 1

- Construct the Euclidean distance matrix for these three cities and their amenities:

City	Hospitals	Clinics	Pharmacies
B	0	4	8
C	0	1	1
D	1	0	0

- What do you notice?

Manhattan Distance

- Manhattan Distance simply sums the absolute differences between values.
 - Values may be positive or negative.
 - In R `dist(A,method="manhattan")`.
- Manhattan Distance (MD) between objects i and h across p dimensions is:

$$MD = \sum_{j=1}^p |(a_{hj} - a_{ij})| \quad (2)$$

- **The distance depends on the scale of the variables.**
- Generally inappropriate for community data.

Example 2

- Construct the Manhattan distance matrix for these three cities and their amenities:

City	Hospitals	Clinics	Pharmacies
B	0	4	8
C	0	1	1
D	1	0	0

- What do you notice?

Jaccard Similarity & Dissimilarity

- The **Jaccard Similarity** measures the *overlap* in attributes between two objects.
- Contingency table:

		Object B	
		Present	Absent
Object A	Present	a	b
	Absent	c	d

- a is the number of common attributes (overlap).
- b number of attributes in A but not in B.
- c number of attributes in B but not in A.
- d attributes that are missing from both (if applicable)

Jaccard Similarity

- The Jaccard similarity (S_J) is calculated as follows:

$$S_J = \frac{a}{a + b + c} \quad (3)$$

- This is a metric measure that is bounded by 0 and 1.
 - Proportion
- Note: *d* is not considered in the calculation.

Jaccard Dissimilarity

- Since the Jaccard similarity is bounded by 1, it is easily converted to a dissimilarity measure by subtraction.
- The Jaccard dissimilarity (D_J) is calculated as follows:

$$D_J = 1 - S_J = \frac{b + c}{a + b + c} \quad (4)$$

- This measures the proportion of attributes that do not overlap in the objects.

Example 3

- Construct the Jaccard similarity contingency table and Jaccard similarity and dissimilarity matrices for these three footballers and their attributes:

Player	Position	Dominant Foot	League	Status
A. Davies	DEF	Left	Bundesliga	Inactive
J. Shaffelburg	MID	Left	MLS	Inactive
J. Waterman	DEF	Right	MLS	Active

- What do you notice?

Minkowski Distance

- Minkowski Distance a generalization of the Euclidean and Manhattan distance measures.
 - Adds an additional parameter p that specifies the root and power to be applied.
 - In R `dist(A,method="minkowski",p=2)`.
- Minkowski Distance ($D_{Minkowski}$) between objects i and h across m dimensions is:

$$D_{Minkowski} = \left(\sum_{j=1}^m |(a_{hj} - a_{ij})|^p \right)^{\frac{1}{p}} \quad (5)$$

- **The distance depends on the scale of the variables.**
- $p = 1$: Manhattan distance.
- $p = 2$: Euclidean distance.
- If $p < 1$ then the measure is not a metric.

Example 4

- Construct Minkowski distance matrices for these three cities and their amenities for the following values of p :

① $p = 1.5$

② $p = 3$

③ $p = 50$

City	Hospitals	Clinics	Pharmacies
B	0	4	8
C	0	1	1
D	1	0	0

- Compare these results to Examples 1 and 2. What do you notice?

Comments on Distance Measures

- We have covered a small fraction of distance measures available for these kinds of analyses.
- I encourage you to explore some of the other distance measures available.
 - Ecological measures and references found in this book.
 - Principal Component Analysis (PCA) are based on Euclidean distances.
- The behaviour of a distance measure is highly influenced by the data adjustments applied prior to its calculation.
- *We may cover and utilize other distance measures throughout this course.*

Next Lecture

- We will discuss more **Multivariate Outlier Detection**.
- Mahalanobis Distance.
- Outlier Detection algorithms for non-normally distributed samples.

Exercise 1

- Use the distance measures that we have covered to determine different distance matrices for this dataset:

Plot	X_1	X_2	X_3	X_4
J	3	1	5	7
K	7	3	4	4
M	4	2	6	5
P	3	3	0	2

- Play around with different values for p .
- What do you notice?

Exercise 2

- Construct the Jaccard similarity contingency table and Jaccard similarity and dissimilarity matrices for these three cities and their amenities:

City	Hospitals	Clinics	Pharmacies	Mental Health Centres
B	0	4	8	1
C	0	1	1	0
D	1	0	0	2

- Hint: The amenities are either present or not present*
- What do you notice?

References & Resources

- 1 Bakker, J. D. (2026) Applied Multivariate Statistics in R
 - 2 Peng R. D. (2020) *Exploratory Data Analysis with R*
 - 3 Timbers T., Campbell T., and Lee M. (2024) *Data science: A first introduction*. Chapman and Hall/CRC
- `dist()`
 - `vegan::vegdist()`
 - `labdsv::dsvdis()`